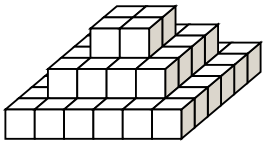
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 39658

Part B: Each question is worth 4 points

1. Sara built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

A. 26 ***B.*** 42 ***C.*** 56 ***D.*** 62 ***E.*** 64

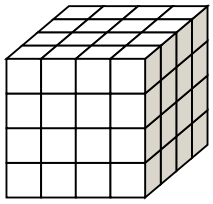
2. Only the even numbers from 2 to 120 are written down: 2, 4, 6, ..., 120.
How many digits are written?

A. 129 ***B.*** 127 ***C.*** 126 ***D.*** 125 ***E.*** 120

3. A counter shows 4843. Every second it goes up by 1.
How many seconds until it shows a number whose 4 digits are all the same?

A. 711 ***B.*** 712 ***C.*** 713 ***D.*** 873 ***E.*** 1823

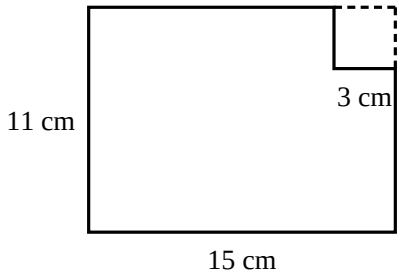
4. A cube made of 4 × 4 × 4 small cubes stands on the table and is painted grey on all sides you can reach: the top and the four sides, but **not** the bottom. Then it is taken apart.



How many small cubes have exactly 2 painted faces?

A. 8 ***B.*** 18 ***C.*** 19 ***D.*** 20 ***E.*** 24

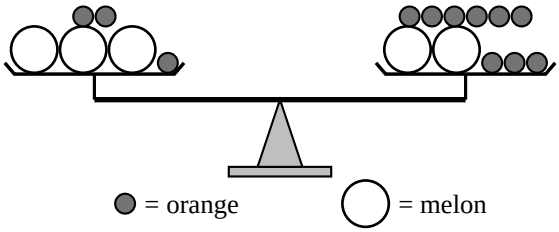
5. A rectangle is 15 cm long and 11 cm wide. A square with side 3 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 40 cm ***B.*** 46 cm ***C.*** 49 cm ***D.*** 52 cm ***E.*** 58 cm

6. The balance is level: 3 melons and 3 oranges on one plate, 2 melons and 9 oranges on the other.



One melon weighs the same as how many oranges?

A. 12 oranges ***B.*** 9 oranges ***C.*** 8 oranges ***D.*** 7 oranges ***E.*** 6 oranges

7. Tom builds a brick 6 cubes long and 4 cubes tall from red and blue cubes. The outside is completely red and the inside is completely blue. Tom used exactly 8 blue cubes.
How many cubes deep is the brick?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

8. Mia was born on February 25, 2010 and Sara on April 1, 2010.
By how many days is Mia older than Sara?

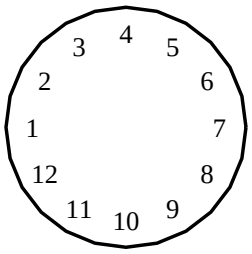
A. 40 ***B.*** 38 ***C.*** 36 ***D.*** 35 ***E.*** 34

9. Emma has 4 different T-shirts, 3 different pairs of trousers and 2 different pairs of shoes. An outfit is one T-shirt and one pair of trousers and one pair of shoes.
How many different outfits can Emma make?

A. 9 ***B.*** 10 ***C.*** 12 ***D.*** 14 ***E.*** 24

Part C: Each question is worth 5 points

10. A round cake with the numbers 1 to 12 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 3 and 4, and between 9 and 10

B. between 4 and 5, and between 11 and 12

C. between 7 and 8, and between 1 and 2

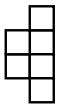
D. between 3 and 4, and between 10 and 11

E. between 4 and 5, and between 9 and 10

11. A bar code is made of 20 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 2 more white bars than wide black bars.
How many narrow black bars are there?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 8

12. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

A.  ***B.*** ***C.*** ***D.*** ***E.***

13. How many different letters are used to write the two words MATHEMATICS and COMPUTER?

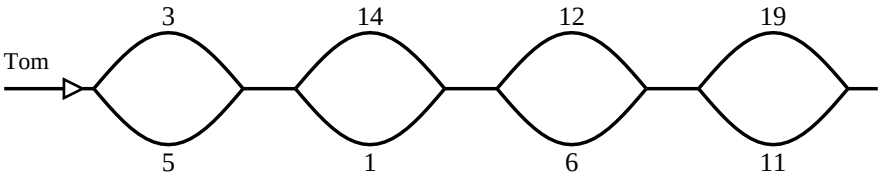
A. 8 ***B.*** 9 ***C.*** 11 ***D.*** 12 ***E.*** 15

14. Lucas, Manon, Omar, Sofia and Chloe each have a different pet: a dog, a parrot, a cat, a goldfish and a canary. Omar has a pet with no legs at all. Sofia doesn't like canaries. Lucas has a pet that barks. Chloe owns a four-legged creature.
Which statement is **not** true?

A. Lucas has a dog ***B.*** Omar has a goldfish ***C.*** Chloe has a cat

D. Manon has a canary ***E.*** Lucas has a cat

15. Tom walks along the road from left to right and adds up the numbers along the way. The total is 27.



Which number did Tom see at the second fork?

A. 1 ***B.*** 6 ***C.*** 12 ***D.*** 14 ***E.*** 19

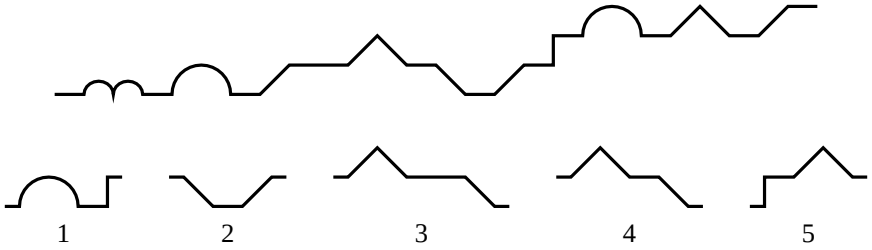
16. Exactly 240 digits are used to write all the page numbers in a book, starting from 1.
How many pages are there?

A. 142 ***B.*** 132 ***C.*** 125 ***D.*** 116 ***E.*** 17

17. Omar has the cards 1, 2, 7 and 9. Using every card exactly once, with +, –, ×, ÷ and brackets, Omar wants to get the biggest possible number.
What is the biggest number Omar can get?

A. 234 ***B.*** 231 ***C.*** 222 ***D.*** 189 ***E.*** 125

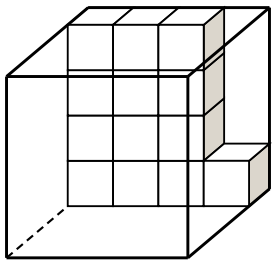
18. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

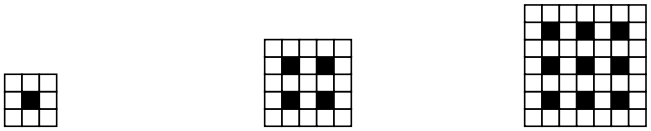
19. Mia puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Mia wants the two bottom layers to be completely full.



How many more cubes are needed for that?

A. 51 ***B.*** 30 ***C.*** 26 ***D.*** 25 ***E.*** 24

20. The diagrams below are the 1st, 2nd and 3rd of a pattern. One diagram further on has exactly 176 white cells. Which one is it?



1st 2nd 3rd

A. the 4th ***B.*** the 5th ***C.*** the 6th ***D.*** the 7th ***E.*** the 11th

Answer Key

1.	**C**	**2.**	**B**	**3.**	**B**	**4.**	**D**	**5.**	**D**	**6.**	**E**
7.	**C**	**8.**	**D**	**9.**	**E**	**10.**	**A**	**11.**	**C**	**12.**	**B**
13.	**D**	**14.**	**E**	**15.**	**A**	**16.**	**D**	**17.**	**D**	**18.**	**B**
19.	**D**	**20.**	**D**								

Solutions

1. C (56) — Count layer by layer, from the bottom: 6 × 6 + 4 × 4 + 2 × 2 = 56. [2002 Q20 · count_pyramid]

2. B (127) — Even one-digit numbers: 4; even two-digit numbers: 45; even three-digit numbers: 11. 4 × 1 + 45 × 2 + 11 × 3 = 127. [2007 Q23 · even_only]

3. B (712) — The numbers with all digits the same are 4444, 5555, ... The next one after 4843 is 5555: 5555 – 4843 = 712. [2002 Q24 · same_digits]

4. D (20) — Top edges without their corners: 4 × 2 = 8 (top and one side). Upright edges without the top corner: 4 × 3 = 12 (two sides; the bottom corners count here because their bottom is not painted). Total 20. [2007 Q17 · not_underneath]

5. D (52 cm) — The two new edges (3 cm and 3 cm) replace the two removed pieces of the outer edge, which were also 3 cm and 3 cm. So the perimeter is the same as the rectangle's: 2 × (15 + 11) = 52 cm. [2002 Q6 · corner_cut]

6. E (6 oranges) — Take 2 melons and 3 oranges off both plates. Left: 1 melon = 6 oranges. [2002 Q9 · both_sides]

7. C (3) — The blue inside is 4 long and 2 tall, so one layer of it has 4 × 2 = 8 cubes. 8 ÷ 8 = 1 layers of blue, plus the red front and back: 3. [2003 Q9 · how_deep]

8. D (35) — February has 28 days, so from February 25 to the end of the month is 3 days, then 32 more days to April 1, 2010: 35 days. [2007 Q12 · days_between]

9. E (24) — Each T-shirt goes with each pair of trousers and each pair of shoes: 4 × 3 × 2 = 24. [2002 Q22 · outfits]

10. A (between 3 and 4, and between 9 and 10) — Total 78, so each piece has 39: 4+5+6+7+8+9 = 39 and 10+11+12+1+2+3 = 39. [2002 Q11 · straight_crack]

11. C (2) — Starting black and ending white, the bars come in pairs: 10 black and 10 white. Wide: 10 – 2 = 8. Narrow: 10 – 8 = 2. [2003 Q17 · ends_differ]

12. B — Two copies have 12 squares, so the rectangle is 1 × 12 or 2 × 6 or 3 × 4. Only two copies of piece B fit together like that. [2007 Q8 · two_copies]

13. D (12) — MATHEMATICS uses 8 different letters and COMPUTER uses 8. C, E, M and T are in both, so count them once: 8 + 8 – 4 = 12. [2007 Q3 · letter_cards]

14. E (Lucas has a cat) — The only possible matching: Lucas – dog; Manon – canary; Omar – goldfish; Sofia – parrot; Chloe – cat. [2002 Q18 · which_not_true]

15. A (1) — Only one route gives 27: 3 + 1 + 12 + 11 = 27. At the second fork: 1. [2007 Q1 · which_branch]

16. D (116) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 240 – 189 = 51 digits are three-digit numbers: 51 ÷ 3 = 17, from 100 to 116. [2003 Q16 · three_digit_pages]

17. D (189) — (1 + 2) × 7 × 9 = 189. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]

18. B (2) — Pieces 2, 4 fit. Not: 1 (it has the right shapes but in the wrong order); 3 (it is a mirror image of a part of the line); 5 (it has a pointed roof where the line has a round dome). [2002 Q7 · how_many_fit]

19. D (25) — Two layers hold 2 × 4 × 4 = 32 cubes. In those two layers there are already 7 cubes (columns count at most 2 each). 32 – 7 = 25. [2007 Q14 · two_layers]

20. D (the 7th) — Try the diagrams one by one: the 7th is 15 × 15 = 225 cells with 49 black, so 176 white. [2007 Q21 · which_diagram]